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Maths puzzles for early finishers

Early Finisher Puzzles

Primary 6

Semester 2

Terms 3 and 4

20 single-page puzzles
Almost every puzzle checks itself
One star and two star pages
Teacher answer key included

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Notes

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Teacher notes (DRAFT)

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What this is

Twenty single-page maths puzzles for Primary 6 pupils who finish their work early. This is Semester 2 of two booklets (Terms 3 and 4). Each page stands alone, so you can hand out one page or keep a tray of pages. The pupil needs no help to start.

Which terms

The pages are grouped by the order in which Primary 6 topics are typically taught: this booklet follows Terms 3 and 4, the other booklet follows Semester 1 (Terms 1 and 2). Schools vary, so check your scheme of work: use a page whenever its topic has been taught, from either booklet.

Stars

One star pages practise a topic as it is usually taught (10 pages). Two star pages use the same topics with two steps or two clues to combine (10 pages).

How the pages check themselves

The boxed strip at the foot of each page tells the pupil how to check. The check shows that something is wrong. It does not give the answer, and it asks the pupil to see whether the answer makes sense. Every answer box and every letter box must be filled: a box left blank gives no letter, and the check cannot see it.

- **Code word.** The answers give letters that spell a word. The strip gives no hint of the word. Each word can be spelt from its letters in one way only, so answers in the wrong boxes never make another word, and one or two wrong answers very rarely do. A box left empty gives no letter and is not seen by the check.
- **Colour by code.** Each answer is found in a key of five numbers and the part is coloured with that crayon. The pupil counts the parts of each colour and compares with the box on the page; matching parts must share a colour.
- **Route.** The line must reach FINISH through the stated number of boxes, and every box on it must follow the rule on the page.
- **Letters in an order, a grid or a choice.** Each number, card, picture or choice carries a letter; the letters, read in the order asked, must spell a word. Where one item is left out, its letter is a decoy that is not part of the word.
- **Loop.** The numbers on touching tiles must match all the way round the loop.

Marking is not needed. To spot-check, look only at the check strip.

How the checks were tested

A script tried every single wrong answer and every wrong arrangement on every page. Each check rejected at least 95% of them. A wrong answer that spells any of about 207 000 English words of two to nine letters counts as not rejected. Two wrong answers at once were not all tested.

Teacher notes: using the pages (DRAFT)

Using the pages

- **Time.** Most pages take 5 to 15 minutes; the colour-by-code pages (S2-12, S2-20) take longer because pupils colour a picture. Pupils need their own crayons (five colours: yellow, red, blue, green, orange). For a pencil-only copy the pupil writes Y, R, B, G or O in each part; the tally and the matching parts are checked the same way. Light crayons work best: a dark crayon hides the number in the part on a photocopy.
- **If the check fails.** The pupil goes back, looks for the slip, and asks a friend or you if still stuck.
- **Problem solving.** Some pages (S2-19) practise the kinds of question that appear in school tests, such as average with a changed value, tank and tap stories and questions that mix several topics, rather than one numbered syllabus outcome. The key tags them problem solving.
- **Stars.** One star: one skill, as usually taught. Two stars: the same topics with two steps or two clues to combine.
- **Drawings.** Pictures are original drawings. Angle figures are not drawn to scale and say so; pupils should work out the answer, not measure it. Pictures are not actual size.
- **Notation.** Ratios are written 3 : 5 and answers in simplest form. Where a page uses π it says whether to take 3.14 or 22/7 (22 over 7). Letters such as n, p and a stand for numbers; there are no brackets with letters. Money is written $12.50. Litres are written ℓ and cube centimetres cm^3.
- **Rough paper and calculators.** Give the pupil rough paper for working. Every number is chosen so that the working can be done by hand; a pupil may use a calculator if your school allows it, but the check does not need one.
- **Revision pages.** Pages tagged Primary 6 Foundation revise earlier-year content and are Foundation-track lines. Primary 6 content only is practised on every other page.

Photocopying

All pages are black and white with solid lines, so they copy cleanly. Use normal toner, not a light setting. Each page is one side only. Pupils write on the page: the boxes are large. Use A4 and print at 100% (actual size).

Pictures are not actual size.

Contents: Semester 2

Page	Code	Puzzle	Tier
S2-01	P6-21	Shaded Shapes	1 star
S2-02	P6-22	Perimeter Pick	2 stars
S2-03	P6-23	Cuboid Corners	1 star
S2-04	P6-24	Root Maze	1 star
S2-05	P6-25	Base and Height	2 stars
S2-06	P6-26	Tank Stories	2 stars
S2-07	P6-27	Angles in Figures	1 star
S2-08	P6-28	More Angles	2 stars
S2-09	P6-29	Angle Match	1 star
S2-10	P6-30	Average	1 star
S2-11	P6-31	Missing Value	2 stars
S2-12	P6-32	Colour the Butterfly	2 stars
S2-13	P6-33	Pie Chart	2 stars
S2-14	P6-34	Fraction Order	1 star
S2-15	P6-35	Ratio Stories	2 stars
S2-16	P6-36	Circle Choices	1 star
S2-17	P6-37	Decimal Maze	1 star
S2-18	P6-38	GST and Discount	1 star
S2-19	P6-39	Mixed Bag	2 stars
S2-20	P6-40	Colour the Elephant	2 stars

Pages are independent. The footer code shows the page in this booklet and, in brackets, its number in the full set of 40 (P6-01 to P6-40).

What the booklet covers

Fractions 1.1 dividing a proper fraction by a whole number **S2-14**

Fractions 1.2 dividing a whole number or proper fraction by a proper fraction **S2-14, S2-19, S2-20**

Ratio 1.2 equivalent ratios **S2-15**

Ratio 1.3 dividing a quantity in a given ratio **S2-15, S2-19**

Ratio 1.6 missing term in equivalent ratios **S2-15**

Ratio 1.7 relationship between fraction and ratio **S2-15**

Algebra 1.4 evaluating by substitution **S2-19, S2-20**

Algebra 1.5 simple linear equations **S2-20**

Area and Volume 1.1 area and circumference of a circle **S2-16**

Area and Volume 1.2 semicircle and quarter circle **S2-02, S2-16**

Area and Volume 1.3 composite figures **S2-01, S2-02**

Area and Volume 2.1 a missing dimension of a cuboid **S2-03**

Area and Volume 2.2 the edge of a cube from its volume **S2-05**

Area and Volume 2.3 height from volume and base area **S2-05, S2-06, S2-12**

Area and Volume 2.4 a face area from volume and one dimension **S2-05**

Area and Volume 2.5 square-root and cube-root signs **S2-04, S2-19**

Geometry 1.1 unknown angles in composite figures **S2-07, S2-08, S2-09**

Statistics 1.1 average: total divided by number of data **S2-10, S2-12, S2-19, S2-20**

Statistics 1.2 average, total and number of data **S2-11, S2-12**

Primary 6 Foundation Statistics 1.1 pie charts **S2-13**

Primary 6 Foundation Decimals 1.1 decimals: multiplying and dividing **S2-17**

Primary 6 Foundation Percentage 1.3 and 1.4 percentage part of a whole; discount, GST, interest **S2-13, S2-18, S2-19, S2-20**

Primary 6 Foundation Area and Volume 2.1 to 2.3 volume of liquid in a tank; litres and cm3 **S2-06**

Outcomes practised in the other booklet: Percentage 1.1; Percentage 1.2; Ratio 1.1; Ratio 1.4; Ratio 1.5; Algebra 1.1; Algebra 1.2; Algebra 1.3.